12-1 Algorithm

Point!

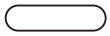
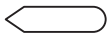
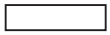
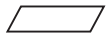
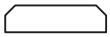
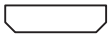
1 Algorithm

(1) ([1]**Algorithm**) : A method or procedure for solving a particular problem.

(2) Flowcharts and activity diagrams : Diagrams that represent algorithms in a visual and comprehensive way.

[1] ([2]**Flowchart**) : Method for illustrating the flow of a single process.

<List of Flowchart Symbols>

Symbol	Name	Meaning
	([3]**Terminal**)	Start/End
	([4]**Display**)	Display on a screen, etc.
	([5]**Data**)	Data input and output
	([6]**Process**)	Operations and other processes
	([7]**Conditional branch**)	Branching according to conditions
	([8]**Repeat**)	Start of a repetition
		End of a repetition
	([9]**Line**)	Flow of data and control

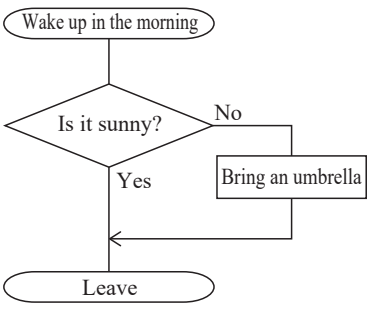
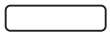
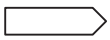


Figure 1: Process flow from waking up to leaving

[2] ([10]**Activity diagram**) : Suitable method for representing parallel process flows.

<List of Symbols in Activity Diagrams>

Symbol	Name
	([11]**Start**)
	([12]**End**)
	([13]**Control**)
	([14]**Transition**)
	([15]**Conditional branch**)
	([16]**Parallel process**)
	([17]**Send**)

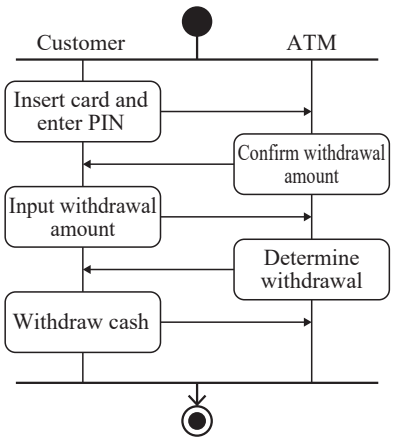
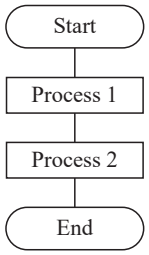
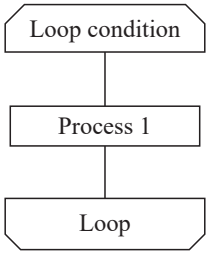
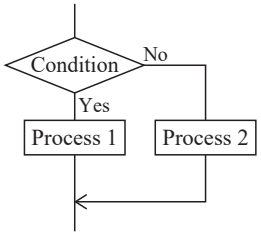


Figure 2: Cash withdrawal process at an ATM

2 Control Structure

Algorithms can essentially be represented using three control structures along with input/output.

([18]**Sequential structure**)	([19]**Repeating structure**)	([20]**Branching structure**)
Executes processes according to the order. 	Repeats the process while the conditions are met. 	Separates processes according to the conditions. 

3 Programming Language

(1) ([21]**Programming language**) : A language used for expressing algorithms in a way that can be understood by a computer.

- Creating a program (source code) using a programming language is referred to as ([22]**programming**).
- Programs are written using programming languages and then translated into a ([23]**machine language**) that can be understood by a computer.

(2) Examples of Programming Languages

[1] ([24]**Python**) : Language used in fields such as artificial intelligence (AI) and statistics, and can be executed with minimal coding.

[2] ([25]**JavaScript**) : Language that can be confirmed only within a web browser, making it optimal for web-related purposes.

[3] ([26]**Scratch**) : A visual programming language developed for educational purposes. Programming is possible using blocks, which is intuitive and easy to understand.

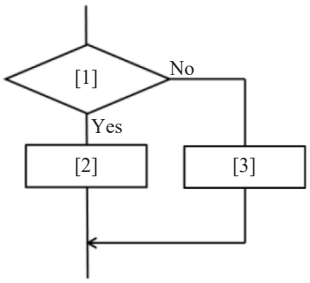
Warm Up

Answer the following questions.

(1) What is the term for describing the order of calculations or the sequence in which things are created?

(2) What is the term for a diagram that visually represents an algorithm and is ideal for showing the flow of parallel processes?

(3) The diagram on the right is a flowchart for pedestrians indicating "Proceed if the signal is green; otherwise, stop." Choose the appropriate option that best fits into the blanks [1] to [3] from the options **A** to **F** below, and answer using the letters.



A The signal is red **B** Signal is yellow
C The signal is not red **D** The signal is not green
E Possible to proceed **F** Stop

(4) What is the term for a control structure like (3)?

(5) What is the programming language that is used in fields like artificial intelligence and statistics, and can be executed with minimal description?

Explanation

(1) Algorithm
(2) Activity diagram
(3) [1] **A** [2] **F** [3] **E**
(4) Branching structure
(5) Python

Try

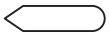
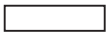
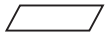
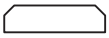
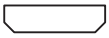
Answer the following questions.

(1) Choose the one that best fits into the blanks [1] to [4] in the following sentences from the options **A** to **F** below, and answer using the letters.

> A method or procedure for solving a particular problem is referred to as an ([1]). A language created to instruct a computer to execute an ([1]) is called ([2]), and creating a program using a ([2]) is called ([3]). In addition, a ([2]) is converted to a ([4]) that can be understood by a computer, and ultimately becomes a set of instructions composed of combinations of 0s and 1s.

A Program **B** Flowchart **C** Algorithm
D Programming **E** Programming language **F** Machine language

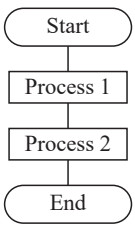
(2) The following table summarizes the symbols used in flowcharts. Choose the option that best fits into the blanks [1] to [8] from the options **A** to **H** below, and answer using the letters.

Symbol	Name	Meaning
	Terminal	([1])
	Display	([2])
	Data	([3])
	Process	([4])
	Conditional branch	([5])
	Repetition	([6])
		([7])
	Line	([8])

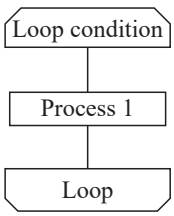
A Start of a repetition
B Start/End
C Flow of data and control
D Branching according to conditions
E End of a repetition
F Display on a screen, etc.
G Data input and output
H Operations and other processes

(3) Choose the one that represents [1] a branching structure and [2] a repeating structure from the following flowcharts **A** to **C**, and answer using the letters.

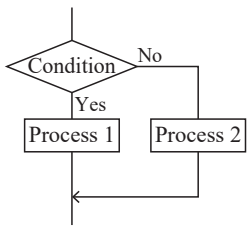
A



B

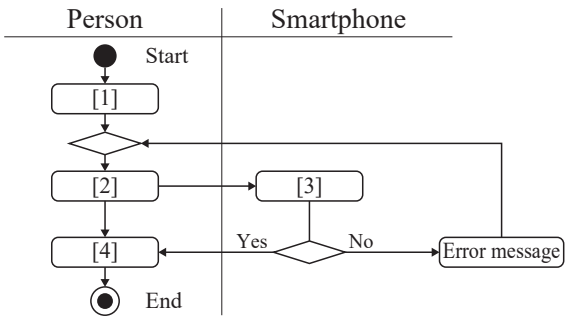


C



(4) The figure on the right is an activity diagram that illustrates the relationship between yourself and a smartphone when unlocking its screen. Choose the option that best fits into the blanks [1] to [4] from the options **A** to **D** below, and answer using the letters.

A Verify password
B Enter password
C Unlock screen
D Screen turns ON



(5) Name the programming language that can be confirmed only within a web browser, making it optimal for web-related purposes.

Exercise

1 Answer the following questions.

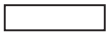
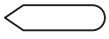
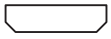
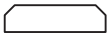
(1) Choose the term that best fits into the blanks [1] to [3] in the following sentences from the options **A** to **F** below, and answer using the letters.

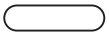
> The order of calculations or the sequence for creating something is referred to as an ([1]). Among the diagrams that visually represent this, the one that is suitable for representing a single process flow is referred to as a ([2]), while the one suitable for representing parallel process flows is referred to as an ([3]).

A Programming **B** Programming language **C** Activity diagram
D Source code **E** Algorithm **F** Flowchart

(2) Choose the symbols used in flowcharts that correspond to the meanings [1] to [4] from the options **A** to **F** below, and answer using the letters.

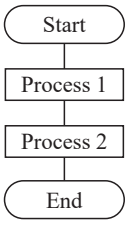
[1] Start/End [2] Branching according to conditions
[3] Start of a repetition [4] Data input and output

A  **B**  **C**  **D** 

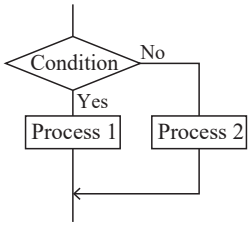
E  **F** 

(3) From the following flowcharts, choose the one from **A** to **C** that best represents a flowchart depicting the case where "If the product is in stock, purchase it; otherwise, stop shopping." Answer with the appropriate letter.

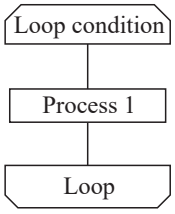
A



B

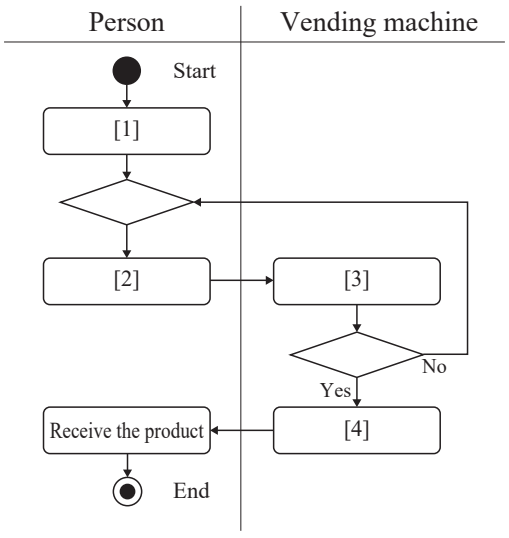


C



(4) The figure on the right is an activity diagram showing the relationship between yourself and a vending machine when purchasing a product. Choose the option that best fits into the blanks [1] to [4] from the options **A** to **D** below, and answer using the letters.

A Insert money
B Money is counted
C Press button to purchase the product
D Product is discharged



(5) Choose the visual programming language developed for beginners and children from the options **A** to **D**, and answer using the letter.

A Python **B** Scratch
C JavaScript **D** BASIC